

CSA0925 – PROGRAMMING WITH JAVA

CO3 – AT – 3

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Q1. Banking Transaction System – Exception Handling
CODE:

```
import java .util.*;
class Main{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
try{
int balance=sc.nextInt();
int amount=sc.nextInt();
if(amount<=0){
throw new IllegalArgumentException("withdraw amount.");
}
if(amount>balance){
throw new InsufficientBalanceException("Insufficient.");
}
balance=balance-amount;
```

```
System.out.println("Withdrawl Successful.");
System.out.println("Remaning Balance = "+ balance);
}
catch(InsufficientBalanceException e){
System.out.println(e.getMessage());
}
catch(IllegalArgumentException e){
System.out.println(e.getMessage());
}
catch(Exception e){
System.out.println("Invalid input.");
}
finally{
System.out.println("Transaction complete.");
}
}
}

class InsufficientBalanceException extends Exception{
InsufficientBalanceException(String message){
super(message);
}
}
```

Q2. Hospital Patient Registration – User-Defined Exception CODE:

```
import java.util.*;

public class Main{

    public static void main(String[] args){

        Scanner sc=new Scanner(System.in);

        try{

            String name=sc.next();

            int age=sc.nextInt();

            double temperature=sc.nextDouble();

            if(age<18|| age >100){

                throw new InvalidAgeException("Invalid age.");

            }

            if(temperature<95.0|| temperature>104.0){

                throw new InvalidTemperatureException("Invalid

                temperature.");

            }

            System.out.println("patient registered successfully");

        }

        catch(InvalidAgeException e){

            System.out.println("InvalidAgeException");

        }

        catch(InvalidTemperatureException e){
```

```

System.out.println("InvalidTemperatureException");
}
finally{
System.out.println("Registration completed.");
}
}
}
class InvalidAgeException extends Exception{
InvalidAgeException(String message){
super(message);
}
}
class InvalidTemperatureException extends Exception{
InvalidTemperatureException(String message){
super(message);
}
}

```

Q3. E-Commerce Order Processing – Built-in Exceptions
CODE:

```

import java.util.*;
public class Main{
public static void main(String args[]){
Scanner sc=new Scanner (System.in);
String[]products={"Laptop","phone","Headphones"};

```

```
double[] prices={50000,20000,3000};
try{
int index=sc.nextInt();
int quantity=sc.nextInt();
if(quantity<=0){
throw new IllegalArgumentException("Invalid quantity.");
}
double total=prices[index]*quantity;
System.out.println("total Cost="+ total);
}
catch(ArrayIndexOutOfBoundsException e){
System.out.println("ArrayIndexOutOfBoundsException
handled.");
}
catch(InputMismatchException e){
System.out.println("InputMismatchedException handled.");
}
catch(IllegalArgumentException e){
System.out.println("Invalid quantity");
}
}
}
```

Q3. E-Commerce Order Processing – Built-in Exceptions

CODE:

```
import java.util.*;

public class Main{

    public static void main(String args[]){

        Scanner sc=new Scanner (System.in);

        String[]products={"Laptop","phone","Headphones"};

        double[] prices={50000,20000,3000};

        try{

            int index=sc.nextInt();

            int quantity=sc.nextInt();

            if(quantity<=0){

                throw new IllegalArgumentException("Invalid quantity.");

            }

            double total=prices[index]*quantity;

            System.out.println("total Cost="+ total);

        }

        catch(ArrayIndexOutOfBoundsException e){

            System.out.println("ArrayIndexOutOfBoundException

            handled.");

        }

        catch(InputMismatchException e){
```

```
System.out.println("InputMismatchedException handled.");
}
catch(IllegalArgumentException e){
System.out.println("Invalid quantity");
}
}
}
```

Q4. Online Food Delivery – Multiple Threads and Execution Order

CODE:

```
import java.util.*;
class Main{
    public static void main(String[] args)throws
InterruptedException{
        Scanner sc=new Scanner(System.in);
        String input=sc.nextLine();
        if(sc.hasNextLine())
            input+=" " + sc.nextLine();
        String[] data=input.split(";");
        String order=data[0].split("=")[1].trim();
        String p=data[1].split("=")[1].trim();
```

```

String d=data[2].split("=")[1].trim();
oThread t1=new oThread(order);
pThread t2=new pThread(p);
dThread t3=new dThread(d);
t1.start();
t1.join();
t2.start();
t2.join();
if(!p.equalsIgnoreCase("Failed")){
    t3.start();
    t3.join();
}
else{
    System.out.println("delivery assignment should not
proceed");
}
}
}
class oThread extends Thread{
    String order;
    oThread(String order){
        this.order=order;
    }
}

```



```

    public void run(){
        System.out.println("order Processing: " + order);
    }
}

class pThread extends Thread{
    String p;
    boolean success=true;
    pThread(String p){
        this.p=p;
    }
    public void run(){
        if(p.equalsIgnoreCase("Failed")){
            success=false;
            System.out.println("payment Processing: "+ p);
        }
    }
}

class dThread extends Thread{
    String d;
    dThread(String d){
        this.d=d;
    }
    public void run(){

```

```
System.out.println("delivery Assignment:"+ d);
```

```
}
```

```
}
```

Q5. Hospital Emergency System – Thread Priority

CODE :

```
import java.util.*;
```

```
public class Main{
```

```
public static void main(String[] args){
```

```
EmThread em=new EmThread();
```

```
GeneralThread g= new GeneralThread();
```

```
BThread b=new BThread();
```

```
em.start();
```

```
g.start();
```

```
b.start();
```

```
}
```

```
}class EmThread extends Thread{
```

```
public void run(){
```

```
System.out.println("Emergency Patient processing");
```

```
}
```

```
}
```

```
class GeneralThread extends Thread{
```

```
public void run(){
```

```
System.out.println("General patient processing");  
}  
}  
class BThread extends Thread{  
public void run(){  
System.out.println("Billing processing");  
}  
}
```